

MEASURING THE TRANSPORT OF PERTURBATION RELIABILITY RANKINGS ACROSS BIOLOGICAL CONTEXTS

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ABSTRACT

Single-cell perturbation predictors are often used to prioritize experiments, but a perturbation that is predicted reliably in one context need not remain reliable in another. We study whether the ordering of perturbation-level realized risk transports across biological contexts when the predictor is held fixed. Because finite cell counts can themselves induce apparent rank changes, we define a tie-aware discrepancy that compares pairwise ordering distributions and removes matched within-context sampling variation. In the Frangieh Perturb-CITE-seq dataset (Frangieh et al., 2021), reliability rankings change across control, co-culture and IFN γ conditions under three complementary risk metrics. The changes exceed same-context sampling baselines, become increasingly resolvable with measurement depth, and are reproduced in independent HepG2–Jurkat and iPSC–neuron raw-cell analyses (Nadig et al., 2025). Matched source-only RBF, direct-MLP, latent-MLP and strict five-fold GEARS sensitivities preserve positive transport signs on the complete Frangieh surface, although their within-metric profiles are not identical. Reliability reordering also changes finite top- k experimental short-lists, although its relation to regret is metric-dependent and is not uniquely explained by a single transport scalar. A known-truth comparator audit shows why the finite-measurement correction matters: raw rank disagreement is spuriously positive under a same-context null, whereas matched-null corrections restore calibration. Together, these results treat predictor reliability as a context-dependent ranking object and provide a finite-measurement framework for deciding when apparent reordering is distinguishable from experimental sampling noise.

1 INTRODUCTION

Single-cell perturbation screens make it possible to measure many interventions in parallel, but the space of candidate experiments grows faster than the experimental budget. Pooled Perturb-seq established a route from genetic perturbations to high-dimensional single-cell readouts (Dixit et al., 2016; Adamson et al., 2016); genome-scale screens and harmonized perturbation resources have since expanded both the intervention universe and the range of measurement depths (Replogle et al., 2022; Peidli et al., 2024). A computational predictor is therefore useful not only for estimating a response, but also for deciding which perturbations deserve follow-up.

That decision is an ordering problem. Existing perturbation-prediction studies usually ask whether a model reconstructs an unseen response or generalizes to an unseen perturbation. Representative approaches span latent-variable, optimal-transport, graph-based and foundation-model methods, including scGen, CPA, CellOT, GEARS, scGPT and TxPert (Lotfollahi et al., 2019; 2023; Bunne et al., 2023; Roohani et al., 2024; Cui et al., 2024; Wenkel et al., 2026). Benchmarks broaden this comparison across datasets and metrics (Wei et al., 2026), while direct audits show that complex models need not outperform simple linear baselines under a fixed protocol (Ahlmann-Eltze et al., 2025; Wong et al., 2025). These studies evaluate response fidelity; they do not directly ask whether the identity of reliably predicted perturbations survives a biological context shift.

A ranking shift is not automatically a biological-context effect. With finite numbers of cells, perturbation risks fluctuate, and two observed orders can differ even when the underlying order is not

resolved. Cell number and sequencing depth are therefore part of the inferential problem, not merely technical details (Peidli et al., 2024; Squair et al., 2021). Work on ranking under noisy measurements makes the same point for uncertain and tied orders (Zuk et al., 2007; Sankaran et al., 2025). Our question is complementary to calibration and selective prediction: after accounting for this finite-cell ordering floor, does the relative ordering of realized model risk change between biological contexts?

We answer this question with a source-frozen protocol. Predictions are fixed before target outcomes are observed; context-specific risks are then compared through a tie-aware, measurement-adjusted discrepancy. For each perturbation pair, the estimator compares the distributions of lower, tied and higher risk states across contexts and removes matched within-context sampling variation using an unbiased order-2 U-statistic (Hoeffding, 1948). The primary analysis uses the Frangieh context surface, a same-context pseudoreplicate control, an independent HepG2–Jurkat replication, matched-depth resampling and a finite top- k decision layer.

We make three contributions. First, we formalize reliability transport as a finite-measurement comparison of perturbation-pair ordering distributions rather than mean response fidelity. Second, we show that context-associated reordering exceeds a matched sampling floor, has an explicit measurement boundary, and recurs under an independent context pair and matched nonlinear predictor sensitivity. Third, we connect the identifiable ordering change to finite experimental shortlists while keeping post-hoc transport measurement separate from prospective source-only prediction.

2 RELATED WORK

Perturbation prediction and its evaluation. Single-cell perturbation prediction has progressed from latent-variable and perturbation-aware generative models to optimal-transport, graph-based and foundation-model approaches (Lotfollahi et al., 2019; 2023; Bunne et al., 2023; Roohani et al., 2024; Cui et al., 2024; Theodoris et al., 2023; Wenkel et al., 2026; Xing & Yau, 2025). Current benchmarks emphasize response fidelity, held-out perturbations and held-out contexts, and show that dataset composition and metric choice can change model conclusions (Wei et al., 2026; Schafer et al., 2026; Vinas Torne et al., 2026). These studies compare predicted and observed responses under specified splits and metrics. They do not treat the ordering of perturbation-level realized risk as the object that must transport across contexts.

Ranking under measurement noise. Calibration and selective prediction study confidence, abstention or risk coverage under a fixed task (Guo et al., 2017; Geifman & El-Yaniv, 2017; Ovadia et al., 2019; Liang et al., 2024). Classical rank correlation and top- k list distances provide deterministic comparators (Kendall, 1938; Fagin et al., 2003); uncertainty analyses study rankings induced by noisy measurements and fragile top- k overlap (Zuk et al., 2007), while weighted and tie-aware extensions address unequal rank importance and incomparable items (Vigna, 2015; Sankaran et al., 2025). Classical distances assume an observed ranking, whereas our setting requires distinguishing context-shifted ordering from instability induced by finite repeated measurement. The order-2 estimator uses the standard U-statistic construction of Hoeffding (Hoeffding, 1948).

From ranking to decisions. Domain generalization formalizes what may survive a domain shift, but the present study makes no causal transportability claim (Jalaldoust & Bareinboim, 2024; Zhou et al., 2023). Decision-focused learning distinguishes predictive error from the loss induced by an action (Donti et al., 2017; Wilder et al., 2019; Elmachoub & Grigas, 2022); here the action is a finite top- k perturbation shortlist, not a learned decision policy. Work on reliable single-cell perturbations instead studies whether the experimental perturbation itself is reproducible (Wang et al., 2026). We use decision loss only to ask whether a frozen source reliability order remains useful after biology changes, keeping data reliability, model reliability and action loss as distinct statistical objects.

3 PROBLEM SETTING AND MEASUREMENT-ADJUSTED ORDERING

Let $x_{p,e}^{(r)}$ be the response for perturbation p in context e and raw-cell resampling replicate r . A predictor f_θ is trained only on a source context and produces a fixed vector $\hat{y}_p$ for each held-out perturbation. For metric m , the realized risk is

$$R_{p,e}^{(m,r)} = g_m(\hat{y}_p, y_{p,e}^{(r)}), \quad (1)$$

Fig. 1 Reliability ordering under biological context shift

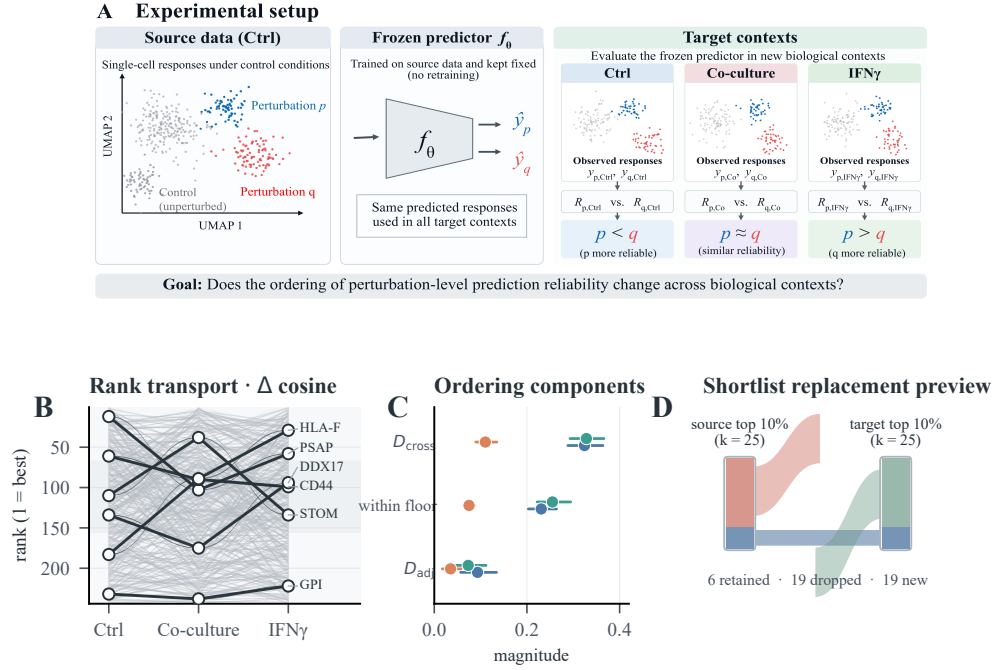


Figure 1: Reliability ordering under biological context shift. The source predictor is frozen before evaluation in the target context. The rank braid retains all 243 background rank trajectories and highlights a deterministic set of source-rank quantile exemplars for legibility. The compact shortlist preview reports 25 perturbations per context: 6 retained, 19 dropped and 19 newly entering the target shortlist. The full 243-label accounting remains in the Fig. 1 source table, while the detailed replacement flow is reserved for Fig. 4A. The decomposition uses the same source-frozen predictor and prespecified evaluation protocol on its declared eligible transfer set. Metric colors are fixed throughout the paper.

where larger values mean worse prediction. We use one minus cosine similarity (delta cosine), one minus Systema centroid accuracy (Vinas Torne et al., 2026), and one minus absolute-effect rank agreement. These risks are evaluated after target outcomes are observed; source-only uncertainty is kept separate from the ordering estimand.

For a pair $a = (p, q)$, define the tie-aware pair state

$$Z_{a,e}^{(m,r)} = \text{sign}\left(R_{p,e}^{(m,r)} - R_{q,e}^{(m,r)}\right) \in \{-1, 0, +1\}. \quad (2)$$

The resampling distribution gives $\pi_{a,e}^{(m)} = [\Pr(Z = -1), \Pr(Z = 0), \Pr(Z = +1)]$. For two contexts e_1, e_2 ,

$$D_{\text{cross}} = \mathbb{E}_a[1 - \pi_{a,e_1}^\top \pi_{a,e_2}], \quad D_{\text{within},e} = \mathbb{E}_a[1 - \|\pi_{a,e}\|_2^2]. \quad (3)$$

The measurement-adjusted discrepancy, equivalently one half of the squared delta-kernel MMD (Gretton et al., 2012), is

$$D_{\text{adj}} = D_{\text{cross}} - \frac{D_{\text{within},e_1} + D_{\text{within},e_2}}{2} = \frac{1}{2} \mathbb{E}_a \|\pi_{a,e_1} - \pi_{a,e_2}\|_2^2 \quad (4)$$

Thus D_{adj} is the squared separation between the two ordering distributions after removing their within-context coincidence terms. It is a transport quantity for relative reliability, rather than an average change in prediction accuracy. For interpretability, write $t_e = \pi_{a,e}(0)$ and $b_e = \pi_{a,e}(+1) -$

$\pi_{a,e}(-1)$. The pairwise squared separation has the exact decomposition

$$\begin{aligned} D_{\text{tie}}(a) &= \frac{3}{4}(t_{e_1} - t_{e_2})^2, \\ D_{\text{adj}}(a) &= D_{\text{tie}}(a) + D_{\text{strength}}(a) + D_{\text{flip}}(a), \quad D_{\text{strength}}(a) = \frac{1}{4}(|b_{e_1}| - |b_{e_2}|)^2, \\ D_{\text{flip}}(a) &= |b_{e_1} b_{e_2}| \mathbb{I}\{b_{e_1} b_{e_2} < 0\}. \end{aligned} \quad (5)$$

For finite measurement, the reported components use a cross-product of independent 15-seed halves while retaining both measurement replicates. The empirical report is therefore $D_{\text{tie}}^{\text{CF}}$ plus a directional-bias component; the nonlinear strength/flip split is retained as population algebra, not presented as an unbiased finite-sample decomposition. The canonical per-seed U-corrected D_{adj} is reconstructed from the merged member-level risks. Finite-measurement component calibration and the comparison between cross-fit and canonical estimates are reported in the Supplement; they are not additional population identities.

Estimand properties. Because each $\pi_{a,e}$ is a probability vector, $0 \leq D_{\text{adj}} \leq 1$. The discrepancy is symmetric in the two contexts and equals zero if and only if the pair-state distributions agree almost everywhere over the declared pair universe. It is therefore a squared divergence; we do not claim a triangle inequality and use “discrepancy” rather than “distance”. The bound applies to the population estimand. Its unbiased finite-sample U-corrected estimate $\hat{D}_{\text{adj}}$ is signed and is not clipped to $[0, 1]$; a negative estimate can occur when the observed pair-state discrepancy is smaller than the finite-replicate correction. In the deterministic tie-free limit, each pair law collapses to a point mass and D_{adj} reduces exactly to the normalized Kendall inversion fraction, $(1 - \tau)/2$. Its distinct role is finite-measurement transport, where pair states are uncertain and the matched within-context component must be estimated rather than ignored. This distinction is not cosmetic. Consider one perturbation pair whose strict ordering probabilities are $(0.9, 0, 0.1)$ in the source and $(0.6, 0, 0.4)$ in the target. Both contexts have the same modal strict order, so Kendall applied to their deterministic mean rankings records no inversion, whereas $D_{\text{adj}}(a) = \frac{1}{2}[(0.9 - 0.6)^2 + (0.1 - 0.4)^2] = 0.09$. The estimand therefore detects transport of ordering reliability even when the mean ranking does not flip. Conversely, an expression-space MMD can be positive when every risk ordering is unchanged; it tests a different population object. Hence D_{adj} is neither renamed Kendall nor a generic distribution-shift score: it is the declared average separation of finite-measurement pair-state laws. For finite resampling sets, the within-context term uses the unbiased order-2 U-statistic (Hoeffding, 1948). A strict direction is declared stable when the 95% equal-tail Beta(1,1) posterior credible interval excludes 0.5, with at least eight non-tie observations supporting the strict comparison; otherwise the pair remains unresolved. This categorical state rule is separate from the 90% intervals reported for effect estimates. At the no-transport boundary the U-statistic is degenerate, so an ordinary percentile bootstrap is not used as a zero-effect test. Detectability is instead assessed from the empirical variance of the 30 independent seed-level U-corrected contributions using a Student- t interval. A complete-seed block bootstrap is retained as a sensitivity analysis. This preserves the registered seed as the independent unit and does not resample the two context arms independently.

The primary Frangieh surface contains 243 shared perturbation labels across three contexts. Fig. 1B selects source-rank quantile exemplars only to keep the complete braid legible, whereas its background ribbons and the Fig. 1 source table retain the full 243-label source-frozen surface. Fig. 1D is a compact decision preview; the detailed shortlist flow appears only in Fig. 4A. Aggregate transport estimates use their declared full eligible tables. The predictor uses a ten-component PCA basis, ridge penalty $\alpha = 0.1$, one global five-fold perturbation split and three independent model members. Matched raw-cell pseudoreplicates use 30 resampling seeds and a minimum of 20 cells per perturbation-context. The independent Nadig analysis uses the exact shared HepG2-Jurkat perturbation set and is not pooled with Frangieh.

3.1 DECISION CONSEQUENCES

For a source context e_s and selection budget k , let

$$S_k(e_s) = \arg \min_{S: |S|=k} \sum_{p \in S} R_{p,e_s} \quad (6)$$

be the source-ranked shortlist. Its target-context loss is $L_{e_t}(S) = k^{-1} \sum_{p \in S} R_{p,e_t}$, and normalized selection regret compares $L_{e_t}(S_k(e_s))$ with the target-oracle shortlist $S_k(e_t)$. Thus the decision

Fig. 2 Context-dependent reliability transport and independent replication

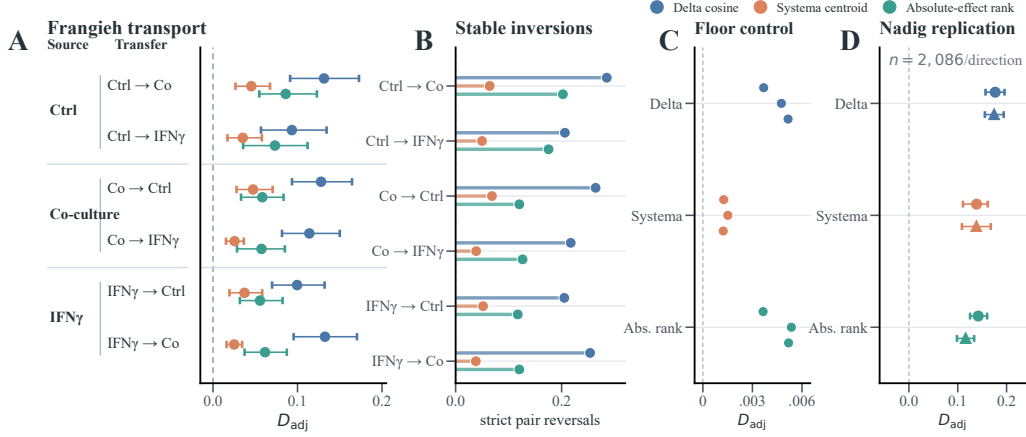


Figure 2: Context-dependent reliability transport. (A) Six directed Frangieh transfers with three metric estimates and 90% intervals. (B) Stable-pair inversion fraction: fraction of pairs stable in both contexts whose strict ordering reverses. (C) Same-context independent pseudoreplicate control, shown as point estimates because it is a floor diagnostic rather than a biological transfer. (D) Independent Nadig HepG2–Jurkat replication in both source directions. Metric color is fixed; in panel D, marker shape identifies the source direction and context is otherwise carried by position and labels.

layer measures the consequence of transporting a source reliability order, rather than evaluating a second predictor.

For the decision layer, the target regret at budget k is defined as

$$\text{Regret}_k(e_s \rightarrow e_t) = L_{e_t}(S_k(e_s)) - L_{e_t}(S_k(e_t)), \quad (7)$$

where the second term is the target-oracle loss. Let $S_k^{\text{worst}}(e_t)$ be the deterministic k -item set with largest target risk and let $L_{e_t}(\mathcal{U})$ be the mean target risk over the eligible universe. We report two explicitly distinct normalizations,

$$\text{Regret}_k^{\text{rand}} = \frac{\text{Regret}_k}{L_{e_t}(\mathcal{U}) - L_{e_t}(S_k(e_t))}, \quad \text{Regret}_k^{\text{worst}} = \frac{\text{Regret}_k}{L_{e_t}(S_k^{\text{worst}}) - L_{e_t}(S_k(e_t))}. \quad (8)$$

The first preserves the random-reference scale used in the primary result; the second is bounded by the deterministic worst shortlist when its denominator is positive.

4 RESULTS

4.1 RELIABILITY ORDERING CHANGES ACROSS BIOLOGICAL CONTEXTS

Figure 1 introduces the object of interest: predictions are held fixed while target biology changes, so movement in the perturbation order cannot be attributed to refitting the predictor. The full Frangieh atlas contains 243 shared labels; Fig. 1B retains all 243 background trajectories and highlights a deterministic set of source-rank quantile exemplars so that individual ribbons remain readable. Preprocessing and replication eligibility are summarized in Appendix A; the complete candidate registry is retained with the repository artifacts. Fig. 1D previews the fixed top-10% shortlist replacement example; the complete label-level risk accounting is retained in the figure source table. The same-context resampling control gives $D_{\text{adj}} = 0.0012 - 0.0054$ across its nine context–metric cells, whereas the matched biological transfers produce larger measurement-adjusted values.

The full-size Ctrl-to-IFN γ transfer illustrates metric dependence. For delta cosine, $D_{\text{cross}} = 0.324175$ and $D_{\text{adj}} = 0.093331$ ($[0.056748, 0.134389]$); for Systema centroid, the corresponding values are 0.110196 and 0.035181 ($[0.017110, 0.057823]$); for absolute-effect rank they are

0.328296 and 0.073300 ([0.035638, 0.111859]). These generated values are read from the same prespecified analysis table used by the figure builder. The conservative joint-floor stress test is retained separately, because it is not an additive component of the measurement-adjusted estimand. The atlas in Fig. 2 retains metric and transfer heterogeneity rather than averaging it away. The finite-sample estimator and tie rule are given in Appendix B.

4.2 STRONG COMPARATOR CALIBRATION IDENTIFIES THE ROLE OF THE CORRECTION

The estimator addresses a finite-measurement target, so a useful comparison is not only against raw rank distances but also against their strongest simple matched-null variants. In a registered 120-condition known-truth simulation, raw Kendall, Spearman and top-10 Jaccard declared transport in 0.952, 0.995 and 1.000 of null conditions, respectively. Label bootstrapping alone did not remove this finite-sample effect. Subtracting a matched within-context null reduced the Kendall false-transport rate to 0.003 and the Jaccard rate to 0.000, while the U-corrected D_{adj} rate was 0.017. At non-zero inversion fractions, detection power was 1.000 for D_{adj} and 0.997 for matched-null Kendall. Thus the result is not that D_{adj} universally dominates every rank statistic: precise measurements make deterministic rank comparators reasonable, whereas D_{adj} supplies one coherent pair-state population estimand with an explicit finite-replicate correction. For the interval itself, direct variance estimation over the 30 independent seed-level U-corrected contributions yielded a null false-transport rate of 0.040 with 0.960 coverage in 2,880 known-truth trials, and 0.015 with 0.985 coverage in 3,000 Frangieh semi-synthetic null trials. These intervals are mildly conservative and replace the anti-conservative hierarchical percentile intervals in every detectability result. The full calibration table, including coverage, bias and the Jaccard comparison, is provided in the Supplement.

The independent Nadig replication provides a different context pair and a larger shared perturbation universe. The macro adjusted excess is 0.176 for delta cosine, 0.138 for Systema centroid and 0.129 for absolute-effect rank, with positive 90% intervals in both source directions. This replication supports the ordering-level phenomenon without being used to tune the Frangieh decision association. Matched-universe sensitivity for this replication is reported in Appendix C, while the complete two-direction full-size surface remains in the figure source table.

A second raw-cell replication reconstructs the complete hierarchy in the Tian iPSC and day-7 neuron CRISPR screens (Tian et al., 2019), rather than using an aggregate rank audit. We froze 26 shared single-gene perturbations, a 1,536-gene panel chosen from source controls only, and a 40-cell budget before evaluating either direction. Five of six direction-metric intervals were positive. From iPSC to neuron, D_{adj} was 0.032 [0.005, 0.060], 0.010 [−0.010, 0.030], and 0.016 [0.004, 0.028] for delta cosine, Systema and absolute-effect rank, respectively; the reverse direction gave 0.049 [0.018, 0.079], 0.042 [0.025, 0.060], and 0.013 [0.001, 0.025]. This asymmetry is itself informative: replication supports context-dependent ordering transport, not a claim that every metric must detect every direction. The complete six-row ledger and source-frozen construction are in Appendix C.

As one matched predictor-family sensitivity, we also materialized a source-only RBF kernel-ridge family on the same Frangieh surface. Its source-only penalty selection required an expanded grid. The matched transport profile had Spearman $\rho = 0.983$ with the primary bilinear-ridge profile (within-metric profile correlations 0.829–1.000; mean row-wise prediction-vector cosine 0.733 in the Supplement), and adjusted-discrepancy point and interval signs agreed on all 18 transfer-metric rows. This is descriptive evidence that the profile is not an immediate artifact of one parameterization, not predictor-family equivalence or broader generalization.

We then ran the official GEARS graph predictor under the same strict information contract. For each Frangieh source context, five label-disjoint folds and all graph construction, fitting and tuning used source cells only; 13 labels absent from the official interaction graph were excluded before splitting, leaving 230 perturbations and 8,229 evaluation genes. The frozen out-of-fold vectors were evaluated by the same 30-seed, two-pseudoreplicate raw-cell bridge. All 18 directed transfer-metric rows retained positive adjusted-discrepancy intervals. Relative to the bilinear profile, GEARS had perfect point-sign agreement but lower magnitude concordance (mean within-metric Spearman $\rho = 0.371$: −0.086, 1.000 and 0.200 for delta cosine, Systema centroid and absolute-effect rank). Thus a modern graph model reproduces the existence of transport while showing that its detailed geometry

Fig. 3 Measurement depth determines when reordering is resolvable

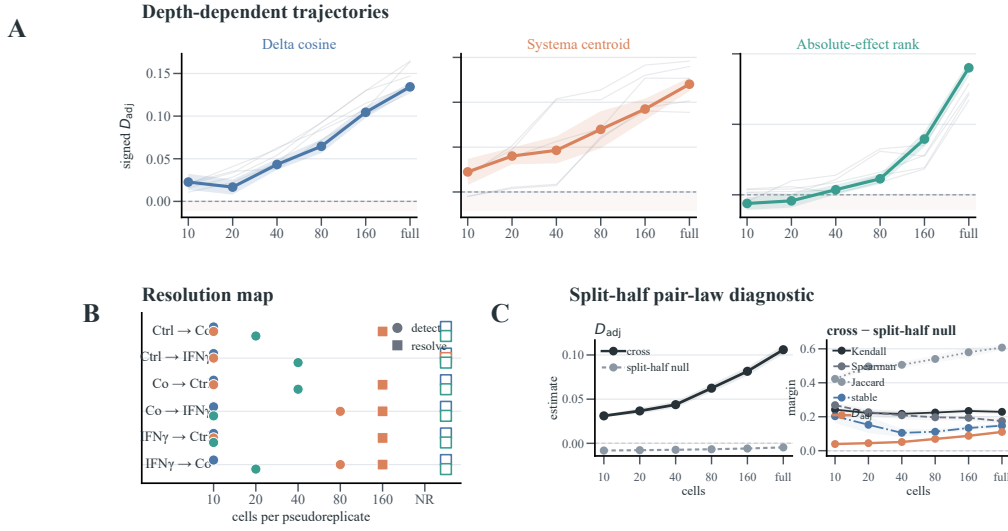


Figure 3: Measurement depth defines an identifiability boundary. (A) Signed depth trajectories for the primary Ctrl–IFN γ contrast with other transfers as pale paths. (B) The six-transfer resolution map; circles mark detectability and squares mark the resolution budget, with open glyphs denoting not reached. (C) Cross-context transport and matched same-context split-half pair-law null margins across depth for the five prespecified comparator families. This is a finite-measurement diagnostic, not the canonical per-seed D_{adj} summary. The full-depth endpoint is the fixed-160 eligibility universe ($n = 139$), not the wider full-size minimum-20 universe used for Figs. 1–2. Signed finite-sample estimates are retained, so null diagnostics can be negative. Detectability and quantitative resolution are separate experimental requirements under the declared resampling protocol.

is predictor- and metric-dependent; it does not establish family equivalence or superiority. The complete surface is reported in Appendix E.

4.3 MEASUREMENT DEPTH DETERMINES WHEN REORDERING IS RESOLVABLE

We next hold the perturbation universe fixed and resample 10, 20, 40, 80 and 160 cells per perturbation–context pseudoreplicate, together with a matched full-depth endpoint. The signed $D_{adj}(n)$ trajectory is retained. The first budget within ten percent of the full-depth value summarizes resolution; detectability requires a positive lower endpoint of the 90% interval. These are metric- and transfer-specific summaries, not a universal cell threshold. We additionally compare each transport statistic with a matched same-context pseudo-context null at every depth, because the canonical within-context measurement term is not itself a same-context D_{adj} estimate.

The trajectories rise at different rates for the three metrics, and several source–contrast cells remain unresolved at the available budgets. Increasing depth therefore improves identifiability without producing a single portable cutoff. This measurement boundary is a prerequisite for interpreting a context-associated ordering difference beyond the measured sampling floor. The matched-null analysis shows positive cross-minus-null margins for every comparator at every depth, but does not support presenting D_{adj} as a universally superior rank statistic; its role is the explicit tie-aware, measurement-adjusted estimand. The depth comparator audit is reported in the Supplementary finite-measurement audit. The depth definitions and all transfer-specific results are in Appendix D.

4.4 CONTEXT-DEPENDENT REORDERING CHANGES SHORTLIST DECISIONS

To connect the ordering to an experimental action, each source context selects the lowest-risk perturbations at budgets $k/n \in \{0.05, 0.10, 0.20, 0.50\}$ and the target context evaluates the selected

Fig. 4 From reordering to experimental regret

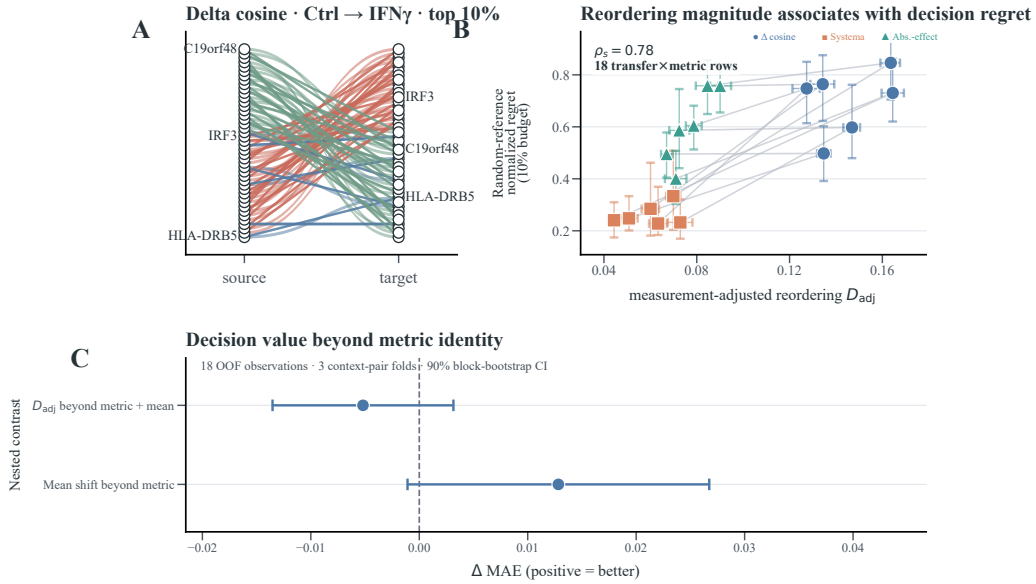


Figure 4: Context-dependent ordering changes experimental priorities. (A) Detailed example of delta-cosine top- k replacement under the Ctrl-to-IFN γ shift (6 of 25 shortlist positions are retained; 19 are dropped and 19 are new). (B) The 18 full-depth transfer-metric observations; marker shape identifies metric and pale connectors identify directed transfer. The pointwise error bars are canonical 90% intervals; the descriptive association and its unordered-context-pair sensitivity are reported in the text, and no fitted line is implied. (C) Unordered-context-pair-held-out incremental prediction of normalized regret beyond metric identity; intervals are block-bootstrap diagnostics.

set. We report target top- k retention and normalized regret under both a random-reference and a bounded worst-oracle normalization.

The primary full-depth link contains six directed source-target surfaces and three metrics, hence 18 descriptive rows. Larger adjusted reordering is aligned with larger normalized regret, with Spearman $\rho = 0.78$; the three unordered context-pair and source-context sensitivities are reported separately. The association is a decision-level summary of the declared surface, not a claim that every predictor or biological system will have the same slope.

An explicit unordered-context-pair leave-one-out comparison separates metric identity, mean-fidelity shift, and ordering transport. Adding the matched mean-risk shift to metric identity reduces OOF MAE by 0.013 ($[-0.001, 0.027]$). Adding D_{adj} beyond metric plus mean shift changes MAE by -0.005 ($[-0.014, 0.003]$); the two nested contrasts are shown separately in Fig. 4. Thus the pooled association is not interpreted as a unique causal contribution of one scalar ordering summary. The corresponding unordered-pair block-bootstrap table is reported in the Supplement; no headline permutation p -value is used for only three blocks.

The shortlist view makes the mechanism concrete: perturbations that occupy the source top- k need not occupy the target top- k , so a source ranking can lose retention and incur regret even when the predictor itself has not changed. The budget-wise association summaries show that the link is not confined to one selection fraction, while its magnitude depends on the transfer and metric. The four-budget, two-normalization robustness summary is reported in Appendix D. A metric-matched mean-risk control, using the same eligible labels and frozen predictions, shows that context-average fidelity and reliability- order transport are empirically non-equivalent (Supplementary Fig. S3).

Additional target-audit and prospective source-only analyses are reported in the Supplement. They are kept separate from the post-hoc transport claim because both require information that is unavailable before target outcomes are measured.

5 DISCUSSION

The central result is that perturbation reliability is an environment-dependent ordering, not a fixed property of a perturbation. With predictions held fixed, matched biological context shifts reordered realized risk beyond a same-context sampling floor, and the direction recurs in both directions of an independent HepG2–Jurkat context pair. The result is therefore an empirical reliability-transport object: it does not claim that every predictor or biological system must reorder in the same way.

This object complements response-prediction benchmarks. scGen, CPA, CellOT, GEARS, scGPT and related studies primarily evaluate response fidelity under specified splits and metrics (Lotfollahi et al., 2019; 2023; Bunne et al., 2023; Roohani et al., 2024; Cui et al., 2024; Wei et al., 2026). Those scores answer whether a model predicts responses well on average; our question is whether its perturbation-level error ordering remains useful after biology changes. Mean fidelity can therefore remain similar while a finite shortlist is replaced. The object is also distinct from experimental perturbation reliability, which concerns whether a measured perturbation effect is reproducible (Wang et al., 2026); we condition on a frozen predictor and make the metric, context and predictor part of the estimand.

Measurement depth supplies an operational boundary for this interpretation. The depth analysis separates detectability of a context-associated ordering change from quantitative resolution of its magnitude. More cells can therefore answer two different experimental questions: whether transport is distinguishable from sampling variation, and whether its estimate is close enough to the full-depth value for quantitative use. The tie-aware support rule and the matched same-context control make that distinction explicit. In the deterministic tie-free limit, D_{adj} reduces to the normalized Kendall inversion fraction; its contribution here is to define the same transport question when pair states are uncertain under finite repeated measurement.

The decision layer shows why the distinction matters. Source-ranked top- k choices lose target retention when reliability moves, and adjusted reordering is descriptively aligned with normalized regret over the six directed surfaces. This is a decision diagnostic, not an end-to-end decision-focused learner (Donti et al., 2017; Wilder et al., 2019; Elmachtoub & Grigas, 2022). The 18 rows are three metrics across three unordered context-pair blocks and both directions, so they are not 18 independent biological replicates. Metric-stratified associations are weaker than the pooled surface, while the matched mean-risk shift provides the only modest OOF improvement (0.013; $[-0.001, 0.027]$) and the additional D_{adj} contrast beyond mean shift does not add held-out value (-0.005 ; $[-0.014, 0.003]$). Thus classical deterministic rank comparators remain appropriate when risks are precisely observed; the paper does not claim that one scalar universally predicts regret.

Scope and testable extensions. The primary decision surface remains anchored to the prespecified bilinear-ridge predictor, three reliability metrics and the Frangieh context design. Source-only RBF, direct-MLP, latent-MLP and official GEARS analyses provide matched predictor-family sensitivities but are not pooled into the primary decision association. Biological transport is independently examined in both the HepG2–Jurkat atlas and a complete iPSC–day-7-neuron raw-cell reconstruction, but neither replication enlarges the Frangieh decision sample. The decision association is therefore still estimated from only three unordered context-pair blocks. Matched source-only RBF and neural-family profiles, the finite-measurement calibration slice, the prospective feature ladder, the active-audit replay and external model controls are reported in the Supplement and are not pooled into the primary claim; the source-only signal should not be read as a pre-experiment oracle. The direct next tests are replication across predictor families, contexts and modalities under the same source-only, label-disjoint and measurement-adjusted contract. The V3 external-validity audit makes this boundary explicit: conventional GEARS cohort runs remain response-prediction controls, whereas the separately executed Frangieh GEARS bridge satisfies the strict source-frozen contract. TxPert is not promoted because a source-matched Frangieh checkpoint and legal prediction vector were unavailable. An outcome-blind Feng multi-donor audit was neutral on the released aggregate LFC surface, so it is reported as a reproducibility boundary rather than a positive biological replication. These exclusions preserve the estimand instead of silently substituting a context-aware model or an incomparable dataset. Context diversity is increasingly recognized as important for virtual-cell generalization (Dibaeinia et al., 2026); the present framework gives such studies a concrete ordering-level quantity to test.

REPRODUCIBILITY STATEMENT

The anonymous core release provides the canonical manifest tables, fixed splits, figure-source products, and commands required to regenerate the four main figures, Supplementary Figs. S1–S5, and the compact tables. Appendices A, B, and H specify the data, estimators, and scientific regeneration path; the transport decomposition, comparator benchmark, incremental decision-value, target-audit, strong-comparator, Feng external-audit and adaptive-allocation ledgers are included as reviewer-facing supplementary artifacts. Auxiliary engineering analyses are retained separately and are not used as manuscript evidence.

CODE AND DATA AVAILABILITY

The anonymized core release is available at <https://github.com/Neabigmo/PTL/tree/iclr2027-anonymous-core>. It contains the analysis and figure-generation code, repository-relative configurations, manuscript sources, publication figures, and compact data products needed to inspect the reported estimates. Raw single-cell matrices are not redistributed. The Frangieh RNA screen and the Nadig HepG2–Jurkat screens can be obtained from the public scPerturb v1.4 Zenodo record; the repository README records the relevant file names and regeneration commands. Users should download the source data directly and follow the original data-set licenses. The Feng multi-donor screen is available from its public Cell Genomics record (Feng et al., 2025); only the outcome-blind compact audit products are included in the anonymous release. AI-assisted tools were used for language editing, literature triage, code organization and figure-script drafting; the authors verified citations, methods and results and are responsible for the final manuscript.

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